Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $fx = \prod_{k=1}^{999} x^2 - 47x + k$, then product of all real roots of $fx = 0$ is

(A) $550!$

(B) $551!$

(C) $552!$

(D) $999!$

Q.2

Let α and β are the roots of equation $ax^2 + bx + c = 0 \quad a \neq 0$. If $1, \quad \alpha + \beta, \quad \alpha\beta$ are in arithmetic progression and $\alpha, \quad 2, \quad \beta$ are in harmonic progression, then the value of $\frac{\alpha^2 + \beta^2 - 2\alpha^2\beta^2}{2\alpha^2 + \beta^2}$ is equal to

(A) 0

(B) 0.5

(C) 1

(D) 1.5

Q.3

If α and β are the roots of the equation $x^2 - 2x + 3 = 0,$ then the sum of roots of the equation having roots as $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ and $\beta^3 - \beta^2 + \beta + 5$ is

(A) 1

(B) 3

(C) 5

(D) 7

Q.4

If the roots of the quadratic equation
$x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$
respectively, then the value of $2 + q - p$ is

(A) 2

(B) 3

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(C) 0

(D) 1

Q.5

Let $x+\frac{1}{x}=1$ and a, b and c are distinct positive integers such that $\left(x^{a}+\frac{1}{x^{a}}\right)+\left(x^{b}+\frac{1}{x^{b}}\right)$ $+\left(x^{c}+\frac{1}{x^{c}}\right)=0$. Then the minimum value of $(a+b+c)$ is

(A) 7

(B) 8

(C) 9

(D) 10

Q.6

A value of b for which the equations

$$x^2+bx-1=0$$
$$x^2+x+b=0$$

have one root in common is

(A) $-\sqrt{2}$

(B) $-i\sqrt{3}$

(C) $i\sqrt{5}$

(D) $\sqrt{2}$

Q.7

If α, β are real and α^2, β^2 are the roots of the equation $a^2x^2-x+1-a^2=0\left(\frac{1}{\sqrt{2}}<a<1\right)$ and $\beta^2 \neq 1$, then $\beta^2=$

(A) a^2

(B) $\frac{1-a^2}{a^2}$

(C) $1-\mathrm{a}^2$

(D) $1+\mathrm{a}^2$

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Q.8

If one root of the equation $(\ell - m)x^2 + \ell x + 1 = 0$ is double the other and ℓ is real, then what is the greatest value of m ?

(A) $-\frac{9}{8}$

(B) $\frac{9}{8}$

(C) $-\frac{8}{9}$

(D) $\frac{8}{9}$

Q.9

If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(A) $3x^2 - 19x + 3 = 0$

(B) $3x^2 + 19x - 3 = 0$

(C) $3x^2 - 19x - 3 = 0$

(D) $x^2 - 5x + 3 = 0$

Q.10

If $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathbf{R}$ and the equations $\mathrm{ax}^2 + \mathrm{bx} + \mathrm{c} = 0$
$a \neq 0$, has real roots α and β satisfying $\alpha < -1$
and $\beta > 1$, then $1 + \frac{\mathrm{c}}{\mathrm{a}} + \left|\frac{\mathrm{b}}{\mathrm{a}}\right|$ is

(A) positive

(B) negative

(C) zero

(D) None

Q.11

If the roots of $ax^2 + bx + c = 0$ are $\sin\alpha$ and $\cos\alpha$ for some α, then which one of the following is correct?

(A) $a^2 + b^2 = 2ac$

(B) $b^2 - c^2 = 2ab$

(C) $b^2 - a^2 = 2ac$

(D) $b^2 + c^2 = 2ab$

Q.12

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If $\lambda \neq \mu$ and $\lambda^2 = 5\lambda - 3, \mu^2 = 5\mu - 3$, then the equation whose roots are $\frac{\lambda}{\mu}$ and $\frac{\mu}{\lambda}$ is

(A) $x^2 - 5x + 3 = 0$

(B) $3x^2 + 19x + 3 = 0$

(C) $3x^2 - 19x + 3 = 0$

(D) $x^2 + 5x - 3 = 0$

Q.13

If $\frac{1}{2-\sqrt{-2}}$ is one of the roots of $ax^2 + bx + c = 0$, where a, b, c are real, then what are the values of a, b, c respectively?

(A) $6, -4, 1$

(B) $4, 6, -1$

(C) $3, -2, 1$

(D) $6, 4, 1$

Q.14

The real roots of the equation $x^2 + 5|x| + 4 = 0$ are

(A) $\{-1, -4\}$

(B) $\{1, 4\}$

(C) $\{-4, 4\}$

(D) None of these

Q.15

If the roots of the equations $px^2 + 2qx + r = 0$ and $qx^2 - 2\sqrt{pr}x + q = 0$ be real, then

(A) $p = q$

(B) $q^2 = pr$

(C) $p^2 = qr$

(D) $r^2 = pr$

Q.16

Consider $f(x) = x^2 - 3x + \mathrm{a} + \frac{1}{a}, a \in R - \{0\}$, such that $f(3) > 0$ and $f(2) \leq 0$. If α and β are the roots of equation $f(x) = 0$ then the value of $\alpha^2 + \beta^2$ is equal to

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(A) greater than 11

(B) less than 5

(C) 5

(D) depends upon α and β cannot be determined.

Q.17

If $y = 2 + \cfrac{1}{4+\cfrac{1}{4+\cfrac{1}{4+\ldots\ldots\ldots\ldots\infty}}}$ then

(A) $y = 6$

(B) $y = 5$

(C) $y = \sqrt{6}$

(D) $y = \sqrt{5}$

Q.18

If $0 < \mathrm{a} < \mathrm{b} < \mathrm{c}$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary then incorrect statement is

(A) $|\alpha| = |\beta|$

(B) $|\alpha| > 1$

(C) $|\beta| < 1$

(D) None of these

Q.19

If z_1, z_2 are the roots of the quadratic equation $az^2 + bz + c = 0$ such that $\mathrm{Im}\,(z_1, z_2) \neq 0$ then (Assume that complex roots are not conjugate to each other)

(A) a, b, c are all real

(B) at least one of a, b, c is real

(C) at least one of a, b, c is imaginary

(D) all of a, b, c are imaginary

Q.20

Suppose the quadratic equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ are such that $\mathrm{p}, \mathrm{q}, \mathrm{r}, \mathrm{s}$ are real and $\mathrm{pr} = 2(\mathrm{q} + \mathrm{s})$. Then

(A) Both the equations always have real roots.

(B) At least one equation always has real roots

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(C) Both the equation always have non real roots

(D) Atleast one equation always has real and equal roots.

Answers & Solutions

Q.1 Answer:

552!

Solution:

Consider $x^2 - 47x + k = 0$

For real roots, $47^2 - 4k \geq 0 \Rightarrow k \leq 552$

$\therefore \quad k = 1, 2, 3 \ldots ., 552$

Product of real roots $= 1 \times 2 \times 3 \times 4 \times \ldots . . \times 552 = 552!$

Q.2 Answer:

1.5

Solution:

$1, \alpha + \beta, \alpha\beta$ **are in A.P.** $\Rightarrow 1, \frac{-b}{a}, \frac{c}{a}$ **are in A.P.**

$\Rightarrow 1 + \frac{c}{a} = \frac{-2b}{a} \Rightarrow a + c + 2b = 0 \ldots 1$

$\frac{1}{\alpha}, \frac{1}{2}, \frac{1}{\beta}$ **are in A.P.** $\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} = 1 \Rightarrow \alpha + \beta = \alpha\beta$

$\Rightarrow \frac{-b}{a} = \frac{c}{a} \Rightarrow b + c = 0 \ldots 2$

From 1 **&** 2 **we get,**

$a = -b = c$

$\Rightarrow \alpha, \beta$ **are roots of equation** $x^2 - x + 1 = 0$

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Now, $\frac{\alpha^2+\beta^2-2\alpha^2\beta^2}{2\alpha^2+\beta^2}=\frac{1}{2}-\frac{(\alpha\beta)^2}{(\alpha+\beta)^2-2\alpha\beta}$

$=\frac{1}{2}-\frac{1^2}{1^2-21}=\frac{1}{2}+1=1.5$

Q.3 Answer:

3

Solution:

Since, $\alpha^2-2\alpha+3=0$

$\Rightarrow \alpha^3-3\alpha^2+5\alpha-2=\alpha\alpha^2-2\alpha+3-\alpha^2-2\alpha+3+1=1$

And $\beta^2-2\beta+3=0$

$\Rightarrow \beta^3-\beta^2+\beta+5=\beta\beta^2-2\beta+3+\beta^2-2\beta+3+2=2$

So, the required equation is

$x^2-2+1x+2.1=0$

$\Rightarrow x^2-3x+2=0$

Sum of roots $=3$

Q.4 Answer:

3

Solution:

Given equation is $x^2+px+q=0$ **Sum of roots** $=\tan 30^\circ+\tan 15^\circ=-\mathrm{p}$
Product of roots $=\tan 30^\circ\cdot\tan 15^\circ=\mathrm{q}$
$\tan 45^\circ=\frac{\tan 30^\circ+\tan 15^\circ}{1-\tan 30^\circ\cdot\tan 15^\circ}=\frac{-p}{1-q}=1$
$\Rightarrow \quad -p=1-q\Rightarrow q-p=1$
$2+q-p=3$

Q.5 Answer:

9

Solution:

$x+\frac{1}{x}=1$
or $x^2-x+1=0$
$\therefore \quad x=\frac{1}{2}\pm i\frac{\sqrt{3}}{2}$

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or $x = e^{\frac{i\pi}{3}}$

$\therefore \quad x^a + x^{-a} = e^{\frac{ia\pi}{3}} + e^{\frac{-ia}{3}} = 2\cos\frac{a\pi}{3}$

Hence, $\cos\frac{a\pi}{3} + \cos\frac{b\pi}{3} + \cos\frac{c\pi}{3} = 0$

$a, b, c \in I \therefore a + b + c|_{\min} = (1 + 3 + 5) = 9$

Q.6 Answer:

$-i\sqrt{3}$

Solution:

Let α be the common root of given equations, then $\alpha^2 + b\alpha - 1 = 0$

and $\alpha^2 + \alpha + b = 0$

Subtracting (2) from (1), we get $(b-1)\alpha - (b+1) = 0$ or $\alpha = \frac{b+1}{b-1}$

Substituting this value of α in equation (1), we get $\left(\frac{b+1}{b-1}\right)^2 + b\left(\frac{b+1}{b-1}\right) - 1 = 0$

or $\quad b^3 + 3b = 0 \Rightarrow b = 0, i\sqrt{3}, -i\sqrt{3}$

Q.7 Answer:

$\frac{1-a^2}{a^2}$

Solution:

$\alpha^2 + \beta^2 = \frac{1}{a^2}$ and $\alpha^2\beta^2 = \frac{1-a^2}{a^2}$

$\Rightarrow \alpha^2 + \beta^2 - 1 = \alpha^2\beta^2 \Rightarrow \left(\alpha^2 - 1\right)\left(\beta^2 - 1\right) = 0$

$\because \beta^2 \neq 1 \Rightarrow \alpha^2 = 1$, so, $\beta^2 = \frac{1-a^2}{a^2}$

Q.8 Answer:

$\frac{9}{8}$

Solution:

Given: $(\ell - m)x^2 + \ell x + 1 = 0$

Let roots be $r, 2r$

$$\text{So } 3r = \frac{-\ell}{(\ell - m)} \text{ and } 2r^2 = \frac{1}{(\ell - m)}$$

$$\Rightarrow \left[\frac{-\ell}{3(\ell - m)}\right]^2 = \frac{1}{2(\ell - m)}$$

$$\Rightarrow 2\ell^2 - 9\ell + 9m = 0$$

if ℓ is real, then $D \geqslant 0$

$81 \geqslant 8 \times 9\ m$

$$\Rightarrow m \leqslant \frac{9}{8}, \text{ Hence maximum value of } m = \frac{9}{8}$$

Q.9 Answer:

$3x^2 - 19x + 3 = 0$

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Solution:

We have $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3; \Rightarrow \alpha \& \beta$ are roots of equation, $x^2 = 5x - 3$ or $x^2 - 5x + 3 = 0$

$\therefore \alpha + \beta = 5$ and $\alpha\beta = 3$

Thus, the equation having $\frac{\alpha}{\beta} \& \frac{\beta}{\alpha}$ as its roots is $x^2 - x\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + \frac{\alpha\beta}{\alpha\beta} = 0 \Rightarrow x^2 - x\left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right) + 1 = 0$ OR $3x^2 - 19x + 3 = 0$

Q.10 Answer:

negative

Solution:

$$
\begin{aligned}
&\text{(b)} \quad \alpha < -1 \text{ . Let } \alpha = -1 - \mathrm{p} \\
&\beta > 1. \text{ Let } \beta = 1 + \mathrm{q}, \mathrm{p} > 0, \mathrm{q} > 0 \\
&\text{Now } 1 + \tfrac{\mathrm{c}}{\mathrm{a}} + \left|\tfrac{\mathrm{b}}{\mathrm{a}}\right| = 1 + \alpha\beta + |-\alpha - \beta| \\
&= 1 + (1+q)(-1-p) + |-1-p+1+q| \\
&= 1 - (1+p+q+pq) + |q-p| \\
&= \begin{cases} -\mathrm{p} - \mathrm{q} - \mathrm{pq} + \mathrm{q} - \mathrm{p} = -2\mathrm{p} - \mathrm{pq} < 0 & \text{if } \mathrm{q} > \mathrm{p} \\ -\mathrm{p} - \mathrm{q} - \mathrm{pq} + \mathrm{p} - \mathrm{q} = -2\mathrm{q} - \mathrm{pq} < 0 & \text{if } \mathrm{q} < \mathrm{p} \end{cases} \\
&\therefore \quad 1 + \tfrac{\mathrm{c}}{\mathrm{a}} + \left|\tfrac{\mathrm{b}}{\mathrm{a}}\right| < 0
\end{aligned}
$$

Q.11 Answer:

$b^2 - a^2 = 2ac$

Solution:

Let $\sin\alpha$ and $\cos\alpha$ be the roots of $ax^2 + bx + c = 0$

Now, $\sin\alpha + \cos\alpha = \frac{-b}{a}$ and $\sin\alpha\cos\alpha = \frac{c}{a}$

Consider $\sin\alpha + \cos\alpha = \frac{-b}{a}$

Squaring both side, $(\sin\alpha + \cos\alpha)^2 = \frac{b^2}{a^2}$

$\Rightarrow \sin^2\alpha + \cos^2\alpha + 2\sin\alpha\cos\alpha = \frac{\mathrm{b}^2}{\mathrm{a}^2}$

$\Rightarrow 1 + \frac{2c}{a} = \frac{b^2}{a^2}$

$\Rightarrow \frac{a+2c}{a} = \frac{b^2}{a^2} \Rightarrow a + 2c = \frac{b^2}{a}$

$\Rightarrow a^2 + 2ac = b^2 \Rightarrow b^2 - a^2 = 2ac$

Q.12 Answer:

$3x^2 - 19x + 3 = 0$

Solution:

λ and μ are the roots of $x^2 = 5x - 3$ or $x^2 - 5x + 3 = 0$

$\therefore \quad \lambda + \mu = 5$ and $\lambda\mu = 3$

$\frac{\lambda}{\mu} + \frac{\mu}{\lambda} = \frac{(\lambda+\mu)^2 - 2\lambda\mu}{\lambda\mu} = \frac{19}{3}$

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$\frac{\lambda}{\mu} \cdot \frac{\mu}{\lambda} = 1$

$\therefore$ **Desired equation is** $x^2 - \frac{19}{3}x + 1 = 0$

or $3x^2 - 19x + 3 = 0$

Q.13 Answer:

$6, -4, 1$

Solution:

Given quadratic equation

is $ax^2 + bx + c = 0$ **whose one root is** $\frac{1}{2-\sqrt{-2}}$

Consider $\frac{1}{2-\sqrt{-2}} = \frac{1}{2-\sqrt{2}i} \times \frac{2+\sqrt{2}i}{2+\sqrt{2}i}$

$= \frac{2+\sqrt{2}i}{4+2} = \frac{2+\sqrt{2}i}{6}$

$\therefore$ **Another root will be** $\frac{2-\sqrt{2}i}{6}$

$(\because$ **complex roots always occurs in pairs** $)$

Thus, sum of roots $= \frac{2+\sqrt{2}i}{6} + \frac{2-2\sqrt{2}i}{6} = \frac{4}{6}$

and product of roots $= \left(\frac{2+\sqrt{2}i}{6}\right)\left(\frac{2-\sqrt{2}i}{6}\right)$

$= \frac{4+2}{36} = \frac{1}{6}$

$\therefore$ **Required equation is** $x^2 - ($ **sum of roots** $)x + ($ **product of roots** $) = 0$

$x^2 - \frac{4}{6}x + \frac{1}{6} = 0$

$\Rightarrow 6x^2 - 4x + 1 = 0$

Thus, the values of a, b, c **are** $6, -4, 1$ **respectively**

Q.14 Answer:

None of these

Solution:

Case 1: $x \geq 0$

$\therefore$ **the equation becomes** $x^2 + 5x + 4 = 0$ **or** $x = -1, -4$ **but** $x \geq 0$

$\therefore$ **both values, non admissible :**

Case $2 : x \leq 0$

The eq ^{n} **becomes** $x^2 - 5x + 4 = 0$ **or** $x = 1, 4$ **both values are non admissible,** $\therefore$ **No real roots.**

Alternatively, since $\mathrm{x}^2 \geq 0; |\mathrm{x}| \geq 0$

$\therefore \mathrm{x}^2 + |\mathrm{x}| + 4 > 0$ **for all** $\mathrm{x} \in \mathbf{R}$

$\therefore \mathrm{x}^2 + |\mathrm{x}| + 4 \neq 0$ **for any** $\mathrm{x} \in \mathbf{R}$

Q.15 Answer:

$q^2 = pr$

Solution:

Consider both equations $px^2 + 2qx + r = 0$ **...(i)**

and $qx^2 - 2\sqrt{pr} \cdot x + q = 0$ **...(ii)**

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Since, both the equations are quadratic and have real roots, therefore from equation (1), we have

$\therefore 4q^2 - 4pr \geq 0$ **(using discriminant)**

$\Rightarrow q^2 \geq pr$ **...(iii)**

and from second equation $4\mathrm{pr} - 4\mathrm{q}^2 \geq 0$

$\Rightarrow pr \geq q^2$ **...(iv)**

From eqs. (iii) and (iv) we get $q^2 = pr$**.**

Q.16 Answer:

5

Solution:

$f(x) = x^2 - 3x + a + \frac{1}{a}; f(3) = 9 - 9 + a + \frac{1}{a} > 0$

$\Rightarrow \quad a + \frac{1}{a} > 0 \Rightarrow a > 0$

$f(2) = 4 - 6 + a + \frac{1}{a} \leq 0 \Rightarrow \quad \frac{a^2 - 2a + 1}{a} \leq 0$

$\Rightarrow \quad \frac{(a-1)^2}{a} \leq 0 \Rightarrow a = 1$

Therefore, $f(x) = x^2 - 3x + 2 = 0$ **has roots 1 and 2.** $\therefore \quad \alpha^2 + \beta^2 = 5$

Q.17 Answer:

$y = \sqrt{5}$

Solution:

$$y - 2 = \frac{1}{4 + \left[\frac{1}{4 + \frac{1}{4 + \ldots\ldots\ldots\ldots\ldots\infty}}\right]} = \frac{1}{4 + (y-2)} = \frac{1}{y+2}$$

$\Rightarrow (y-2)(y+2) = 1 \Rightarrow y^2 - 4 = 1 \Rightarrow y^2 = 5$

$\therefore y = \pm\sqrt{5}$ **since** $y > 0$

$\therefore y = \sqrt{5}$

Q.18 Answer:

$|\beta| < 1$

Solution:

Since the roots are imaginary $\therefore \mathrm{D} < 0$ **and roots occur as conjugate pair, i.e.** $\beta = \bar{\alpha}$

$\therefore |\beta| = |\bar{\alpha}| = |\alpha|$

Also, let $\alpha = \frac{-b + i\sqrt{4ac - b^2}}{2a}$

$\therefore |\alpha| = \sqrt{\frac{b^2}{4a^2} + \frac{4ac - b^2}{4a^2}} = \sqrt{\frac{c}{a}}$

$|\alpha| > 1 (\because \mathrm{c} > \mathrm{a})$

$\therefore |\alpha| = |\beta| > 1$

Q.19 Answer:

at least one of a, b, c is imaginary

Solution:

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Since $az^2 + bz + c = 0$ ...(1)
and z_1, z_2 (roots of (1)) are such that $\text{Im}(z_1 z_2) \neq 0$. Now, z_1 and z_2 are not conjugates of each other Complex roots of (1) are not conjugate of each other
Coefficient a, b, c cannot all be real at least one of a, b, c, is imaginary.

Q.20 Answer:

At least one equation always has real roots

Solution:

Let the discriminant of the equation $\mathrm{x}^2 + \mathrm{px} + \mathrm{q} = 0$ be D_1, then $\mathrm{D}_1 = \mathrm{p}^2 - 4\mathrm{q}$ and the discriminant D_2 of the equation $\mathrm{x}^2 + \mathrm{rx} + \mathrm{s} = 0$ is $\mathrm{D}_2 = \mathrm{r}^2 - 4\,\mathrm{s}$

$\therefore D_1 + D_2 = p^2 + r^2 - 4(q + s) = p^2 + r^2 - 2pr$

[from the given relation]

$\therefore D_1 + D_2 = (p - r)^2 \geq 0$

Clearly at least one of D_1 and D_2 must be non-negative consequently at least one of the equation has real roots.

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